A COHERENT MODULE AGREES WITH ITS PUSHFORWARD OFF A LOCUS OF DEPTH AT LEAST TWO

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 0-2000 granted to this lane by CONTROL 8.83.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- Criterion for an isomorphism via depth and associated points. Lemma lemma-check-isomorphism-via-depth-and-ass (268-280), used at proof lines 787-789. Reduces the claim to two pointwise conditions.
- Weakly associated points of a pushforward along an open immersion. Lemma lemma-weakass-pushforward, used at proof lines 792-794. Places the points outside U outside Ass(j_*(F|_U)).
- Associated points lie in the support. Lemma lemma-weakly-ass-support, used at proof lines 792-794. Supports the same conclusion.

Source reference. Original lemma 0AVM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13690–13690. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0AVN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13691–13691. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05AM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6855–6855. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let X be a locally Noetherian scheme. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. Let $j : U \to X$ be an open subscheme such that for $x \in X \setminus U$ we have $\text{depth}(\mathcal{F}_x) \geq 2$. Then*

$$\mathcal{F} \longrightarrow j_*(\mathcal{F}|_U)$$

is an isomorphism and consequently $\Gamma(X, \mathcal{F}) \to \Gamma(U, \mathcal{F})$ is an isomorphism too.

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. We claim Lemma 0AVM applies to the map displayed in the lemma. Let $x \in X$. If $x \in U$, then the map is an isomorphism on stalks as $j_*(\mathcal{F}|_U)|_U = \mathcal{F}|_U$. If $x \in X \setminus U$, then $x \notin \mathrm{Ass}(j_*(\mathcal{F}|_U))$ (Lemmas 0AVN and 05AM). Since we've assumed $\mathrm{depth}(\mathcal{F}_x) \geq 2$ this finishes the proof. $\square$